

Vibrating antennas and compensation techniques

Research in NATO/RTO/SET 087/RTG 50

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Abstract—Array antennas which are integrated onto structures of aircraft and Unmanned Aerial Vehicles (UAVs) are subject to unsteady aerodynamic loads. Mechanical forces and these aerodynamic loads will cause deformation of the antenna supporting structure. As a consequence, the positions and orientations of the elements of the phased array antenna change. The relative phases of the respective signals feeding the antennas will vary, and as a consequence the antenna radiation pattern is affected: the main beam direction can change and the beam width and/or side lobe levels can increase. The influence of deformations and vibrations will be most significant on array antennas, which are large in terms of wavelength (high gain antennas). The objective of the present paper is to present some applications of such array antennas, and to discuss the effects of deformations and vibrations on the performance of array antennas, and to describe technology to counteract these effects by means of adaptive or synthetic beam forming.^{1,2}

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¹ 1-4244-0525-4/07/\$20.00 ©2007 IEEE.

² IEEEAC paper #1703, Version 7, Updated November 24, 2006

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1. INTRODUCTION

Advanced military aircraft and modern Medium and High Altitude Long Endurance (MALE, HALE) unmanned aerial vehicles (UAVs) will be equipped with structural integrated array antennas to fulfil avionics functions, which are mainly related to radar, Electro-magnetic Counter Measures (ECM) and Communication, Navigation and Identification (CNI). Conformal array antennas can be realised by means of integration of arrays of micro-strip multi-layer antenna elements in the skin of aircraft. When such antenna elements are put on the skin of the aircraft they are subject to steady and unsteady aerodynamic loads. The inertial forces and these aerodynamic loads will cause deformations and vibrations of the total antenna. As a consequence, the positions and slopes of the elements of the array antenna change. The influence of deformations and vibrations will be most significant on array antennas, which are large in terms of wavelength (high gain antennas). An example of such an antenna is an array antenna for side-looking Synthetic Aperture Radar (SAR) mounted on the fuselage of an UAV (see Figure 1) or on a reconnaissance pod of a fighter aircraft. Such an array antenna consists of a group of antenna elements in which the relative phases of the respective signals feeding the antenna elements are varied in such a way that the effective radiation pattern of the array is

reinforced in the side-looking direction. A SAR application for reconnaissance tasks requires in general highly directional antennas with a narrow beam and low side-lobe levels. Any error in the main beam direction, beam-width or an increase of the side-lobe levels caused by structural vibrations will have a significant effect on the final image produced by the SAR.

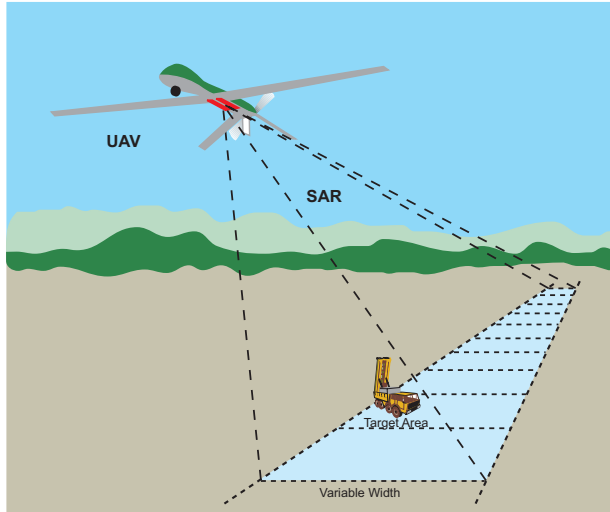


Figure 1 UAV equipped with side-looking synthetic aperture radar

In the NATO Task Group RTO/SET 087/RTG 50 multidisciplinary research has been performed by the research institutes ONERA (France), FGAN (Research Institute for High Frequency Physics and Radar, Germany) and National Aerospace Laboratory NLR (the Netherlands). Furthermore, THALES Airborne Systems (France) and the Air Force Research Laboratories (USA) have contributed to the specification of large aerospace antennas, which are subject to deformations and vibrations.

The NATO Task Group RTO/SET 087/RTG 50 has addressed the following topics:

- Assessment of levels of vibrations of aerospace structures supporting phased array antennas (specification of antennas of interest and external loads, simulations using aero-elastic methods).
- Evaluation of performance of vibrating aerospace antennas (by means of electromagnetic modelling of deformed antennas and signal processing studies).
- Compensation techniques (active vibration control, array shape measurements and electronic compensation, auto-calibration techniques).

- Demonstrators (linear array of patch antenna elements on a vibrating plate, linear interferometer array fixed on a cantilever wing type mock-up).

The objective of the present paper is to present the major achievements of the work done in the NATO Task Group RTO/SET 087/RTG050.

The outline of this paper is as follows. In section 2 we present three typical applications of array antennas which are subject to deformations and vibrations. In section 3 we present a research tool to estimate the levels of vibrations of aerospace structures which are supporting array antennas. In section 4 the effects of deformations and vibrations on large antennas are discussed. In section 5 the capabilities of compensation techniques counteracting the effects of vibrations are presented. In section 6 results of a demonstrator antenna on a vibrating structure in an experimental setup are presented. Finally some conclusions are drawn in section 7.

2. APPLICATIONS

Large interferometer array on STEMME motor glider

ONERA considers the integration of a large interferometer array on a STEMME motor glider. This antenna is formed by two linear sub-arrays placed along the wing and the fuselage (see Figure 2). The goal is to locate an emitter on the ground, with the highest possible accuracy. To achieve this, the array size must be large compared to the wavelength of the received signals. However, the deformations/vibrations of the structure of the plane go against the high expected angular accuracy. Indeed, they generate changes of the positions of the antenna elements. This causes an error in the interferometer processing, which becomes more harmful as the array needs to be more accurate.



Figure 2 Illustration of interferometer array on motor glider (red points are antenna elements)

Taking into account all the difficulties which to this ambitious project must face, ONERA conducted a theoretical study based on computer simulations to evaluate the project viability. Furthermore, experimental tests with a mock-up in anechoic chamber have been performed (see section 6).

Transatlantic Cooperative Advanced Radar

NATO considers the development of an airborne ground surveillance radar system, known as the NATO Alliance Ground Surveillance System (AGS). Six nations in NATO are involved in the development of a Transatlantic Cooperative Advanced Radar. The TCAR is an advanced side-looking wide area surveillance (WAS) multi-mode radar. The radar is characterized by long range, simultaneous SAR and MTI-functions, and an active electronically scanned array (AESA) X-band antenna. The TCAR will be installed on a strut on top of the fuselage of an Airbus 321 aircraft (see Figure 3). The dimensions of the X-band antenna are about 6 by 1 meter, consisting of 6 panels of 1 by 1 meter with sub-arrays (see Figure 4). The supporting strut will likely be not sufficiently stiff, with the consequence that the panels of the X-band array antenna will start to vibrate. In order to achieve the MTI clutter rejection ratio ($>50\text{dB}$) irrespective to their frequencies the amplitudes of the vibration modes must be less than some tens μm . It is expected that the amplitudes will be larger. The frequencies and amplitudes of these vibrations have to be assessed to evaluate the consequences on clutter rejection. Motion sensors on the sub-arrays could be needed to compensate for the distortion by post-processing.

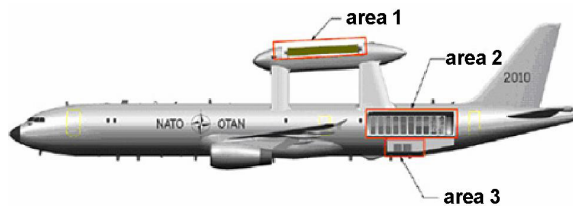


Figure 3 Aircraft with TCAR antenna on top of fuselage

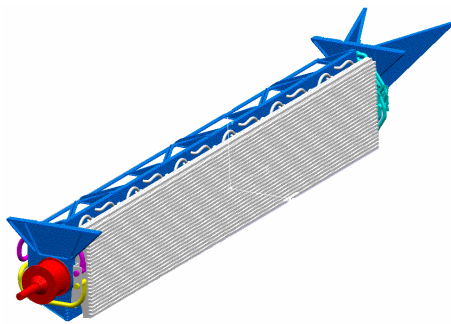


Figure 4 Artist impression of TCAR X-band antenna in Area 1 of Figure 3

Innovative Space-based radar Antenna Technology

Air Force Research Laboratories (AFRL) is involved in the development of the Innovative Space-based radar Antenna Technology (ISAT) program of the Defense Advanced Research Projects Agency (DARPA). This program focuses on developing systems to deploy extremely large (up to 100 meters) electronically scanning radar antennas flying at medium earth orbit altitudes. The primary goal is supporting the war fighter through development of tactical grade, ground moving target indication (GMTI) capability. The ISAT flight experiment includes advanced antenna architectures and structures, as well as advanced metrology and calibration concepts for larger radar antennas. Lightweight materials are being used for the supporting structure of the radar antenna. Due to thermal loads the structure can deform. The vibrations of the ISAT structure have been characterized in the frame of NATO TG50.

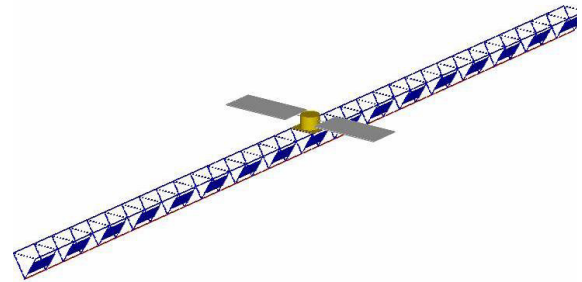


Figure 5 Artist impression of ISAT

3. ASSESSMENT OF LEVELS OF VIBRATION

Through the years the analysis of the dynamics of aircraft has been based on separate calculations of vibration characteristics of the aircraft structure and unsteady airloads. The vibration modes and reduced frequencies are computed by means of elasto-mechanical modelling. The unsteady airloads are then calculated for a range of reduced frequencies and vibration modes by unsteady aerodynamic analysis methods. Subsequently, the deformation of the aircraft structure (i.e. the amplitudes) is computed by combining the vibration modes and the unsteady airloads for representative flight conditions (e.g. a gust input). The generalised modal deflection system of the aircraft structure relates the amplitudes of the vibration modes to the unsteady dynamic loads.

The elasto-mechanical calculations require the availability of structural finite element data of the aircraft as well as computational methods. For the aircraft the vibration modes are distinguished for global modes (related to low frequency oscillations) and local modes (related to high frequency oscillations). For the computation of the global vibration

modes a structural model is required of the relevant part of the aircraft (e.g. fuselage, wing or vertical tail), where many details of the structure are not taken into account. However, for the computation of local vibration modes a more detailed model is required for a part of the aircraft structure where the antenna will be located. In this case, the finite element model should at least include the mass and stiffness properties of the spars, stiffeners and the skin plate.

For the analysis of phased array antennas on vibrating structures the effects of both rigid and flexible motions of the aircraft surface are of interest. Rigid vibrations will change the direction of the main beam of the antenna, while the side lobes are not affected. This effect can be taken into account in the signal processing. As a consequence of flexible vibrations of the structure, the positions of the elements of a phased array will change during a cycle of oscillation. For side looking radar antennas this has a negative effect on the radiation pattern: beam width and side lobe levels will increase.

The prediction of airloads on oscillating aircraft structures in subsonic and transonic flow requires the application of unsteady aerodynamic computational methods. Such methods have been developed since the seventies. For relatively low frequency oscillations, global unsteady airloads can be computed by means of so-called doublet lattice methods (these methods assume that the unsteady aerodynamic disturbances are small relative due to the steady aerodynamic flow state of a stationary aircraft). For relatively high frequency oscillations, local unsteady airloads can be calculated by means of so-called piston theory. This theory is based on a local relation (i.e. locally at the antenna surface) between pressure and normal component of the fluid velocity.

The aero-elastic analysis requires that the static and dynamic deformations of the aircraft geometry be determined, subject to aerodynamic and external forces acting on the aircraft. The geometrical state of the aircraft can be described by the superposition of:

- The jig shape (i.e. the shape of the aircraft in unloaded state),
- The deformation due to steady aerodynamic forces (i.e. the change from the jig shape to the design shape under cruise conditions) and
- The dynamic deformations which are related to displacements belonging to some prescribed vibration modes.

The amplitudes of the external forces are usually prescribed in relation to representative flight conditions.

A study has been performed to estimate the deformation levels of wings of various aircraft types due to occurrences of gusts. The study is based on the Von Karman gust spectrum. This spectrum is based on empirical data and depends on parameters representing: altitude, a reference rms gust speed and a turbulence scale. It defines the number of occurrences of gust being expected within a given period of time for the specified parameters.

For many commercially available aircraft types a detailed and calibrated aero-elastic model is available only to the design office of the manufacturer. In order to overcome this deficiency of knowledge NLR has developed in-house the computational tool GENNAS. This tool is capable of generating a very rough aero-elastic model based on data that is available in the public domain (e.g. Janes All The Worlds Aircraft or the internet). The generated model consists of a MSC/NASTRAN beam representation of the aircraft structure and MSC/NASTRAN aerodynamic panels representing the lifting surfaces. The GENNAS tool automatically generates an analysis model based on a very limited number of parameters. The aero dynamic loads are based on doublet lattice theory. Figure 6 presents a typical model. For a given Von Karman spectrum, other NLR tools can compute wing deflection amplitudes due to gust responses and can relate these to their occurrence per flight hour.

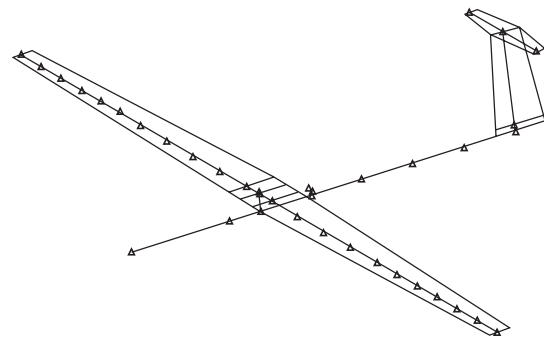


Figure 6 Simplified structural, aerodynamic and mass model for STEMME generated by GENNAS

The results of a typical analysis for the STEMME motor glider are presented in Figure 7 and Figure 8. Figure 7 presents the undeformed front view of the simplified aircraft model in blue circles. For the dots representing the wing the vertical displacements relative to the wing root due to a gust response are plotted as red circles. The plot represents displacement levels that are exceeded once per flight hour. Figure 8 presents the exceedance curve for the wing tip. Note that higher levels are exceeded less frequently. Results of this model were compared to other reference data. The results of the simplified model are correct in their order of magnitude. Note that deflections are in the order of 75 cm once per hour.

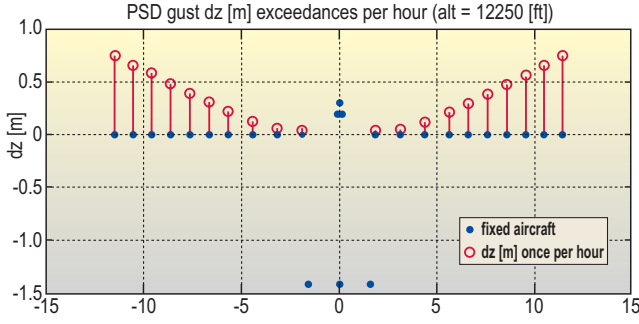


Figure 7 Assessment of wing deflection amplitudes for STEMME motor glider using GENNAS model

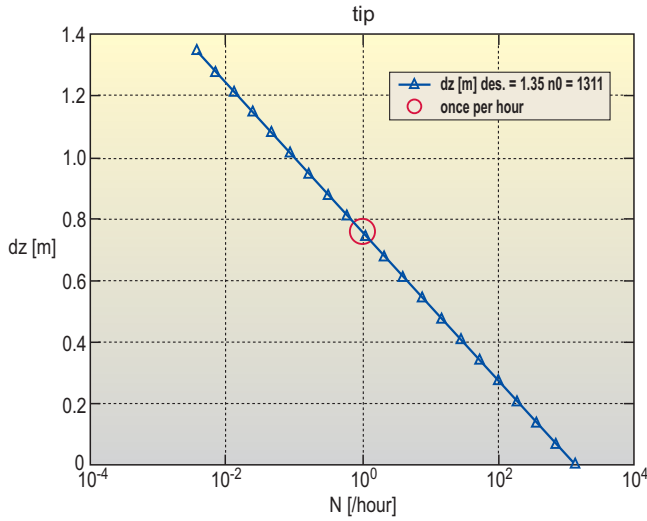


Figure 8 Assessment of wing tip deflections with exceeding per hour, predicted by the GENNAS model

The GENNAS tool has also been used to predict the vibration levels of the wings of a civilian airliner and the Global Hawk, using only open source data. Many dimensions can be measured from drawings. Other data, such as mass distributions, can be estimated from tabular data and statistical data available from the internet or handbooks on preliminary aircraft design.

4. EFFECTS OF VIBRATIONS ON ANTENNAS

In the present paper a generic array antenna on a vibrating plate (see Figure 9) is considered as a typical test case. The array antenna has eight embedded patch elements in receive mode. The influence of vibrations is investigated for a radar aperture antenna and a generic interferometer array antenna. The latter one contains smart signal processing algorithms to identify direction of arrival (DOA) of signals. Here the MUSIC (Multiple Signal Classification) algorithm is applied to finding the spatial spectrum of the antenna array.

In computer simulations the deformation and vibration of the plate (of Figure 9) can be represented by superposition of the bending and first vibration mode. It is assumed that the plate is subject to a forced vibration at $x = 0$. Let $\alpha(t)$ represent the motion of the plate at $x = 0$, which reads $\alpha(t) = w_0 \cos(\omega t)$, with w_0 the amplitude and ω the angular frequency of the vibration.



Figure 9 Vibrating plate with embedded patch antennas

Then, the deformation of the plate can be modelled as

$$z(x, t) = Z_0(x) + \alpha(t) + q_1(t)Z_1(x) \quad (1)$$

where Z_0 represents the bending mode and Z_1 the first vibration mode. Detailed expressions for Z_0 , Z_1 and $q_1(t)$ have been presented in Ref. [1].

The phased array antenna on the vibrating plate is modelled as a receive antenna, which is illuminated by a plane incident wave with direction $\hat{k}^i = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ and wave number $k = 2\pi / \lambda_{RF}$ with λ_{RF} the wavelength of the RF signal. The patch antennas are modelled as isotropic radiators. The centres of the patches are given by the coordinates $\vec{r}_j = (x_j, y_j, z_j)$, $j=1, \dots, 8$. Then, the electric field at patch j becomes

$$\vec{E}_j = E_o \hat{e}^j e^{-ik(x_j \sin \theta \cos \phi + y_j \sin \theta \sin \phi + z_j \cos \theta)} \quad (2)$$

The centres of the patches are on the vibrating plate. As a consequence, the phase in equation (2) is non-stationary due to vibrations. The received signals of the patches will vary in time.

In order to obtain a beam forming array antenna the excitations (i.e. phases and amplitudes) of the elements have to be specified and the separate signals have to be

combined. Then, the total electric field of the array antenna becomes

$$\vec{E}_{tot} = \sum_j A_j \vec{E}_j \quad (3)$$

where A_j are the excitations of the antenna elements. The modulus of A_j (related to the amplitude) is used for tapering and the argument of A_j (related to the phase of the amplitude) for beam steering.

Some effects of vibration on the received signal of the array antenna are shown in Figure 10 and Figure 11. The undisturbed array antenna is located in the plane $z=0$. Neither tapering nor beam steering is applied; hence the excitations $A_j=1, \forall j$ in equation (3). The centres of the patches on the vibrating plate are located at $\vec{r}_j = (x_j, 0, z_j)$, with $x_j = d + (j-1)0.6\lambda_{RF}$. Here, the distance d between the first patch and the reference patch is 10 cm and $\lambda_{RF} = 16.5$ cm, which corresponds with an RF frequency of about 1.8 GHz. Furthermore, it is assumed that the array is illuminated by a plane wave with incident direction $\hat{k}^i = (\sin\theta, 0, \cos\theta)$; hence the azimuth incident angle $\varphi=0$. For $\theta = \pm 30^\circ, \pm 15^\circ, 0^\circ$ the variation of the amplitude of the co-polar component of the total electric field in equation (3) is displayed in Figure 10. The pink line corresponds with illumination normal to the plane (i.e. $\theta=0$). The other lines correspond with illumination angles $\theta = \pm 15^\circ$ and $\theta = \pm 30^\circ$. It can be observed that the received strength varies with time.

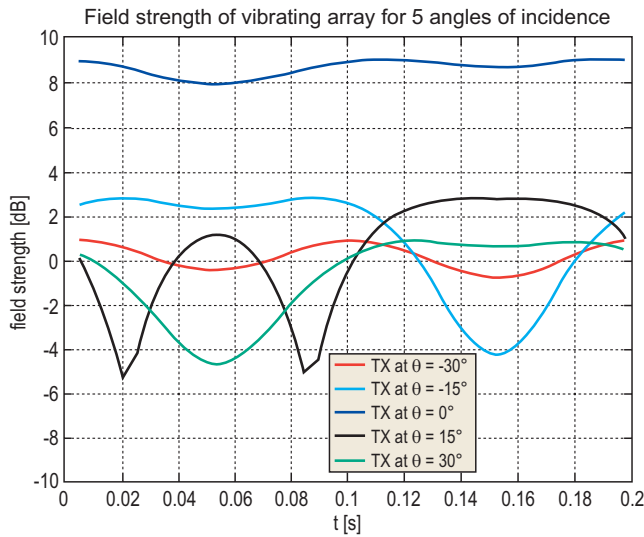


Figure 10 Output of phased array antenna on vibrating plane, for five angles of illumination

This effect is also observed from Figure 11, where the radiation pattern of the array on the vibrating plate is displayed for three different time steps during one cycle of oscillation. This figure shows that the direction of the main beam changes during the oscillation.

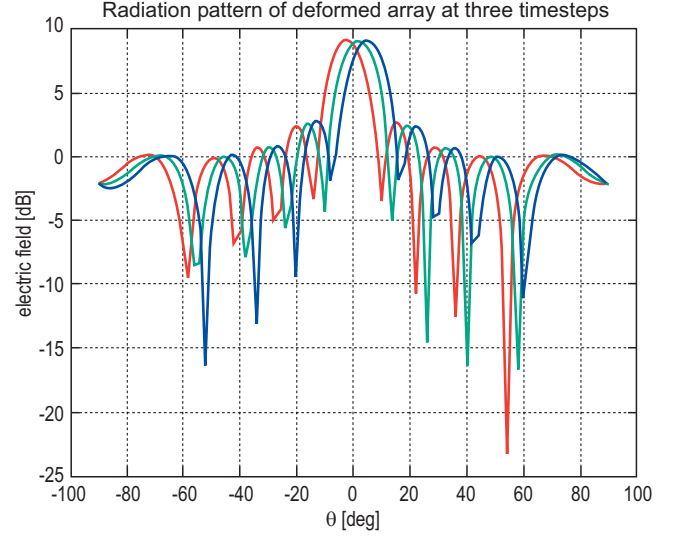


Figure 11 Radiation pattern of phased array on vibrating plane at three different time steps during one cycle of oscillation

The processing of multiple signals received on an array of antenna elements can provide information about number of signals, direction of arrival (DOA) of the signals, and polarisation of the signal. Special applications are interferometers and monopulse direction finding. In Ref. [5] the MUSIC algorithm has been described to estimate DOA using an array of antenna elements. In this section the effects of vibration of the array antenna on estimating DOA are investigated. For the unperturbed plate and the steadily bent plate the MUSIC spectra are displayed in Figure 12 and Figure 13. The peak in Figure 12 reveals that one signal is arriving at an incident angle of $\theta = 38$ degrees. In case the plate is bending, the centres of the patches have a small deflection in z -direction. As a consequence, the phases of the incident signal have changed. Although the incident signal itself is exactly equal to the unperturbed case, it appears from Figure 13 that the signal arrives now from a direction $\theta = 36$ degrees.

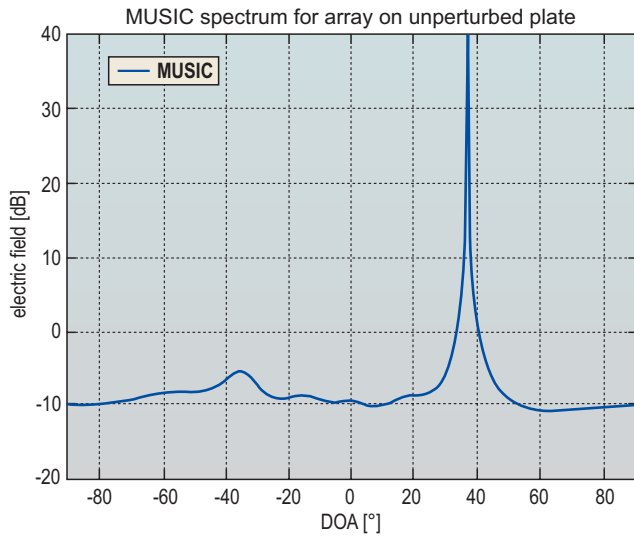


Figure 12 MUSIC power spectrum for linear 8x1 array of patch elements on unperturbed plate

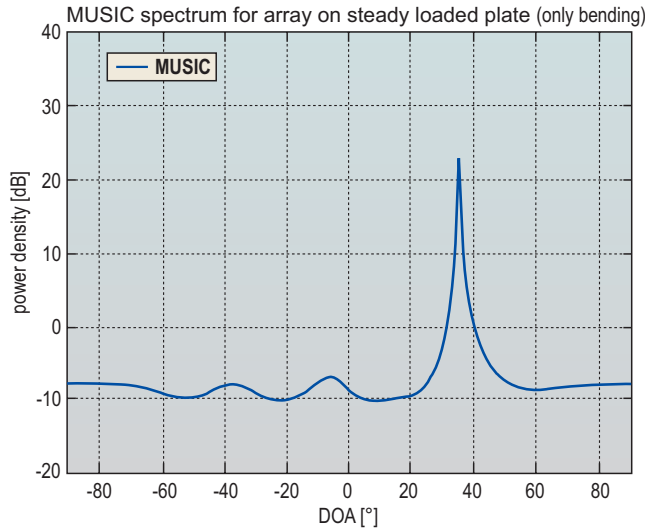


Figure 13 MUSIC power spectrum for linear 8x1 array of patch elements on steadily bent plate

The calculation of the MUSIC spectrum for the array on the vibrating plate requires the specification of the number of snapshots T in the matrix X of received signals and the covariance matrix. One cycle of oscillation of the vibrating plate is sampled by 40 time steps. First, the received data of four subsequent time steps is used to fill the matrix with received signals. Then, the MUSIC spectrum is computed at four different time steps during one cycle of oscillation. The results have been displayed in Figure 14. This figure reveals that the direction of arrival of the signal changes. Next, all sampled data during one cycle of oscillation is used to compute the MUSIC spectrum. Hence, the number of snapshots in matrix X of received signals is now equal to 40. The result is shown Figure 15. The peak in the MUSIC

spectrum appears again at $\theta = 38$ degrees, which corresponds with the arrival direction of the unperturbed plate of Figure 12. However, the peak in Figure 15 is less pronounced than in Figure 12. The signal processing of all snapshots during one cycle of oscillation results in some averaging of data, and spreading of the peak in the MUSIC spectrum.

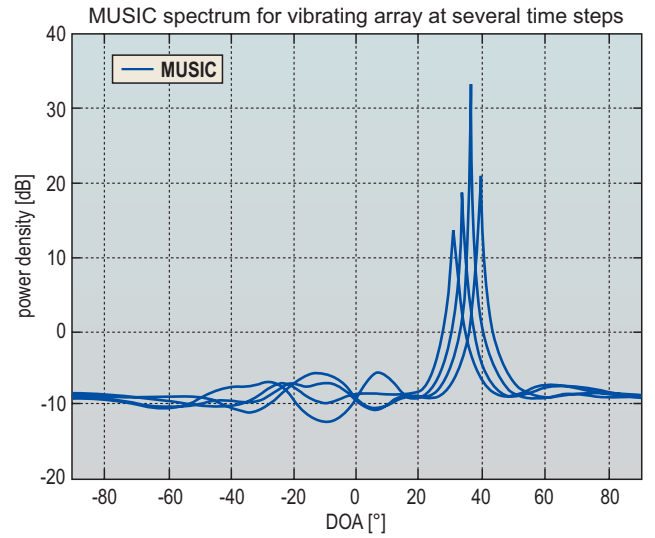


Figure 14 MUSIC power spectrum for linear 8x1 array of patch elements on vibrating plate (4 snapshots at four time steps)

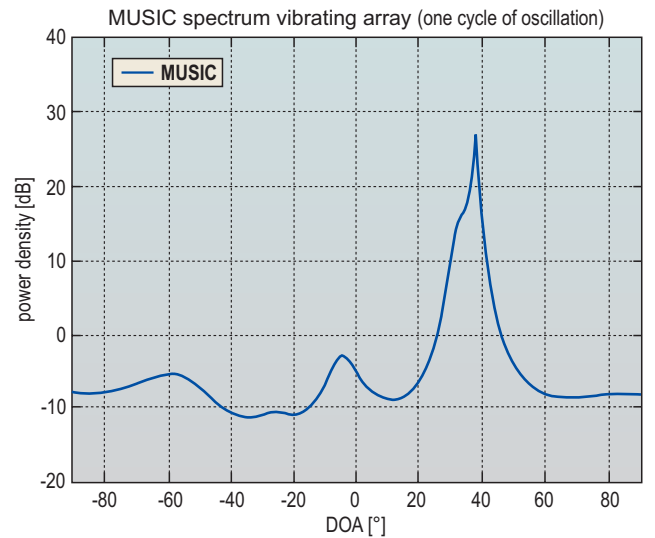


Figure 15 MUSIC power spectrum for linear 8x1 array of patch elements on vibrating plate (40 snapshots during one cycle of oscillation)

5. COMPENSATION TECHNIQUES

From the previous sections it can be concluded that changes in phase (due to out-of plane deformations and vibrations) have to be compensated. The phase difference between a patch antenna on the deformed plate and the unperturbed plate at $z = 0$ is given by

$$\Delta\vartheta_j = -k\{(x_j - x_j'')\sin\theta\cos\varphi + z(x_j, t_0 + n\Delta t)\cos\theta\} \quad (4)$$

In order to establish the phase difference it is required to measure instantaneously the displacement z and the direction of arrival (θ, φ) of the incident wave. When only the displacement z can be assessed an approximate phase correction follows from taking an incident field perpendicular to the plate $z = 0$, i.e. with incident direction $(\theta, \varphi) = (0, 0)$. Then, the approximate phase correction becomes

$$\Delta\vartheta_j \approx -kz(x_j, t_0 + n\Delta t) \quad (5)$$

The phase errors in the signal can be compensated by modifying instantaneously the argument of the excitations in equation (3) as follows

$$A_j := A_j e^{-i\Delta\vartheta_j} \quad (6)$$

When these excitations are applied in equation (3), the beam is in fact instantaneously steered.

Figure 16 displays the output of the phased array antenna (at 1.8 GHz) on the vibrating plate with these synthetic phase corrections. A comparison of Figure 10 (displaying the total without any beam steering) and Figure 16 reveals that the total signal becomes already almost stationary for incoming plane waves with incident angles between $\pm 30^\circ$. This is also observed from Figure 17, where the radiation pattern of the array on the vibrating plate is displayed for three different time steps during one cycle of oscillation. This figure shows that the direction of the main beam is now stationary, contrary to Figure 11. When Figure 11 and Figure 17 are compared, then only for grazing incident waves the radiation patterns are different. This can be explained since the $\cos(\theta)$ term in equation (4) has been taken constant (to one) in approximation (5).

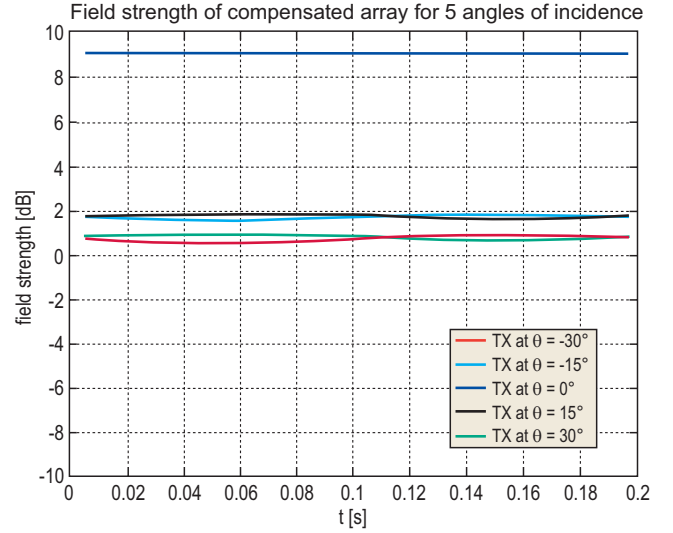


Figure 16 Output of phased array antenna with approximate synthetic compensation, for five angles of illumination

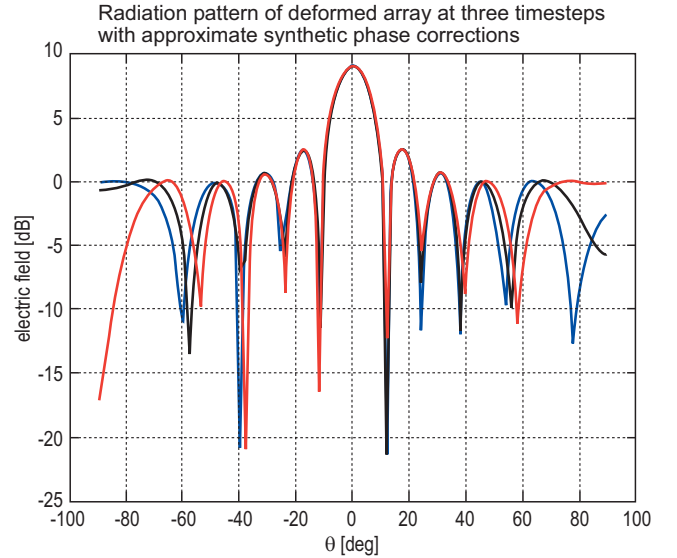


Figure 17 Radiation pattern of phased array on vibrating plane at three different time steps with approximate synthetic phase compensation

The synthetic phase correction in approximation (5) assumes that the displacement z can be instantaneously assessed, which is not a straightforward task. An alternative technology involves measuring the phase differences with respect to the undisturbed steady state of the plate. This can be performed approximately as described below. Consider a reference patch antenna on a stable part of the aero-structure. Let the co-ordinates of the reference antenna be given by $\vec{r}_{ref} = (x_{ref}, y_{ref}, z_{ref})$. Let $\vec{r}_{j,u}$ be the co-ordinates of the centre of patch j of the rigid aluminium plate in the plane $z = 0$. Then, the phase difference between patch j and the reference antenna becomes

$$\Delta\vartheta_{j,u} = -k(\vec{r}_{j,u} - \vec{r}_{ref}) \bullet \hat{k}^i \quad (7)$$

Let $\vec{r}_{j,n}$ be the co-ordinates of the centre of patch j of the vibrating plate at time step t_n . For the patches on the vibrating plate the phase difference between patch j and the reference antenna reads

$$\Delta\vartheta_{j,n} = -k(\vec{r}_{j,n} - \vec{r}_{ref}) \bullet \hat{k}^i \quad (8)$$

The difference between $\Delta\vartheta_{j,n}$ and $\Delta\vartheta_{j,u}$ is obtained by subtracting equation (7) from equation (8)

$$\Delta\vartheta_{j,n} - \Delta\vartheta_{j,u} = -k(\vec{r}_{j,n} - \vec{r}_{j,u}) \bullet \hat{k}^i \quad (9)$$

Note that the right hand side of equation (9) is the phase difference between patch j at the vibrating plate (at time step t_n) and patch j at the unperturbed rigid aluminium plate. Hence, when $\Delta\vartheta_{j,n}$ and $\Delta\vartheta_{j,u}$ can both be measured, an estimate for the right hand side of equation (9) follows from subtracting $\Delta\vartheta_{j,n}$ and $\Delta\vartheta_{j,u}$.

This type of phase correction has been applied to the array antenna on the vibrating plate (see Figure 9). The patch antenna close to the fixture (in the centre of the plate) has been taken as reference element. The phase difference between patch j and the reference patch is assessed by using the AD8302 Integrated Circuit of Analog Devices that detects the RF/IF gain and phase in relation to a reference signal. The product description of this circuit has been given in Ref. [4].

6. DEMONSTRATOR ANTENNA

Description of experiment

The experimentation is described on the following pictures:

- the mock-up represents a wing (laid out vertically), and the fuselage (horizontal), on reduced scale;
- it is placed in front of 3 transmitting sources, each one centred on 3 different frequencies in the Ka band;
- two linear sub-arrays of 5 patch antennas are laid out on the wing and the fuselage, in reception mode only ;

The wing is the only one suffering deformations. It is equipped with a set of mechanical components, allowing on one hand the activation of these deformations, and in the other hand the measurements and estimate of the shape of the wing and the antennas positions.

The configuration in the anechoic chamber (see Figure 18) corresponds to the scene in real flight (horizontal motor glider, emitters on the ground) after a 90° rotation. Hence, the wing is laid out vertically and the line of sight to the “motorglider-emitters” is almost horizontal.

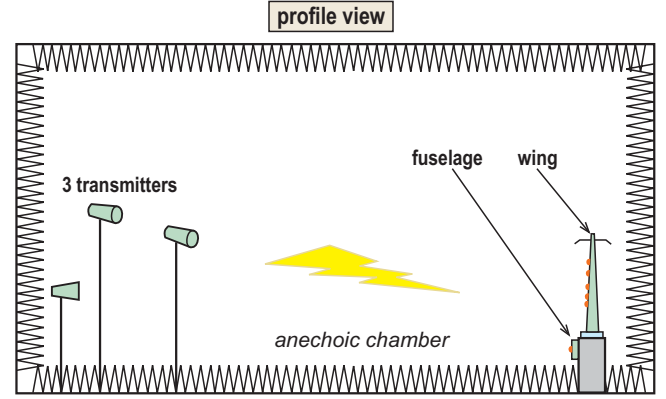


Figure 18 Artist view of the experiment, carried out in anechoic chamber

Figure 19 shows a front view of the mock-up, equipped with the 2x5 patch antennas (supplied with flexible microstrip lines) and with a set of mechanical components.

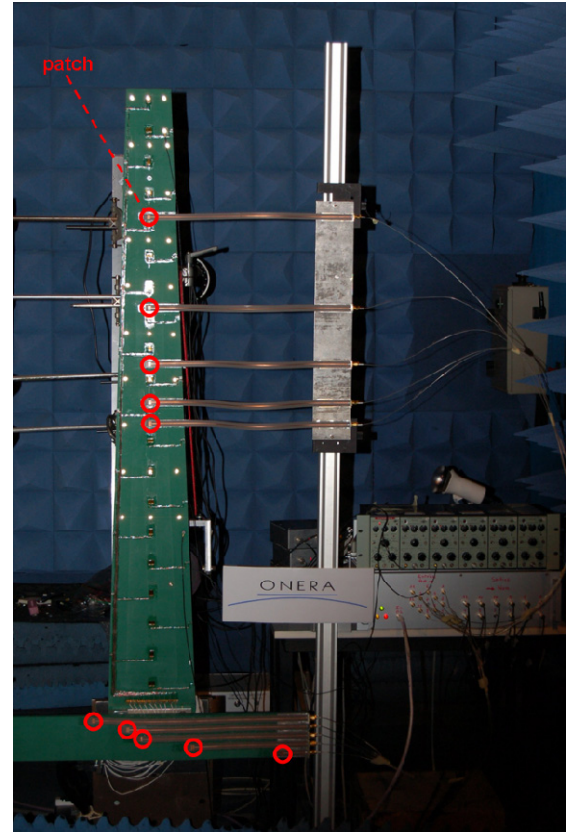


Figure 19 Front view of the experimental interferometer array

Figure 20 shows the three emitters in the anechoic chamber of which the location has to be determined by means of interferometry processing using the experimental array as displayed in Figure 19.

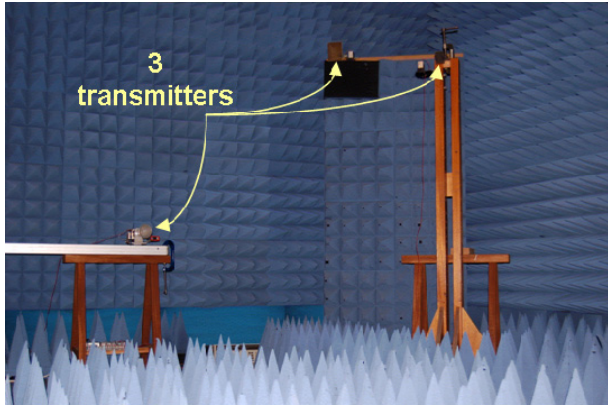


Figure 20 Three transmitters in experimental set-up

Principle of auto focus interferometry

For a perfectly known and not deformed antenna array, the interferometry processing consists in minimising the distance between a measurement and the goniometry table (computed “off line” and stored in memory). To do this, we have:

- to compute “off line” the goniometry table, which associates a set of N_c sensors phases to a given Direction Of Arrival (or DOA) θ , and to store it in memory ;
- to assign each sample of received signal to a position in the table, thus giving a direct estimate of the DOA $\hat{\theta}$. This assignment is done by the minimisation of the distance between the set of the N_c sensors signal phases and the whole sets stored in the table.

For a deformed array, the processing requires additional stages:

- estimate of the positions of antennas, knowing some signals and/or mechanical measurements;
- “on line” calculation of the corrective term of the goniometry table;
- Minimisation of the distance, to locate the unknown sources.

The general principle of the antenna shape estimation method developed by ONERA is summed up in Figure 21, where θ_s is the DOA of the signal; z_0 is the initial position

of the antennas, on z-axis (axe of deformation); $\hat{z}$ is the estimated position

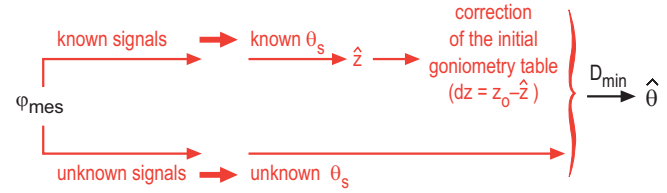


Figure 21 Main steps of interferometer location, with autofocus based on known signals

In the experimentation, the only deformation undergone by the mock-up is along axis Oz, for the sub-array of the wing. Therefore, we perform only a 1D-estimation along axis Oz, and a computation of the displacement along the sub-array axis. Note that although the sub-array of the wing is the only one being deformed, we estimated the positions of the two sub-arrays (wing and fuselage).

Performance without deformation

The performance of the interferometer without any deformation of the wing has been checked, by comparing the estimated accuracy of location to the theoretical one.

Figure 22 shows the 2D location “on the ground” of the three sources, estimated during time (group of points). The ellipses drawn in continuous line correspond to the σ -bounds, and those in dotted line to the 3σ -bounds. For Gaussian errors as it's the case here (except when angular ambiguities occur), these bounds contain respectively more than 68% and 99% of points. The red ellipses correspond to *theoretical* values, taking into account dimensions of the interferometer, the direction of arrival of signals and their estimated S/B. The blue ellipses indicate the *estimated* bounds of the group of points. The results of the three sources S1, S2 and S3 are displayed respectively from the top to the bottom of each figure.

The picture shows a very good agreement between the estimated and theoretical values (respectively the blue and red ellipses). Thus, we can consider the experimental interferometer as efficient. The small differences in size, between estimated and theoretical ellipses, are due to the fact that the latter uses the estimated S/B, which have been probably over-estimated. To confirm this point, we checked by simulation that a perfectly known S/B gives close results between theory and estimations.

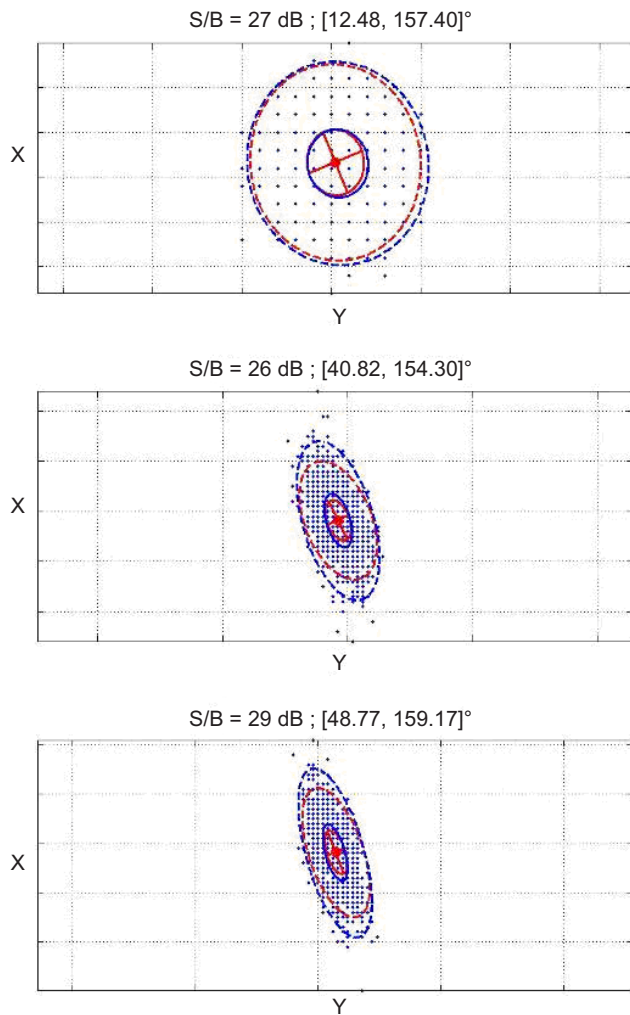


Figure 22 2D locations on ground of the 3 sources (reference case, without deformation)

Performance with steady deformation

Figure 23 presents one example of the static case, with level of deformation $\lambda/2$. Compared to the reference case (Figure 22) on the left), this case shows that the level of deformation is relatively high for the array since it generates an important bias in the interferometer location processing, which challenges the performances of the array. However, using photographs of the Stemme motorglider in flight, we evaluated the deformation of the array of about 1.8λ . The effects on the motorglider thus will be more penalizing.

Performance with deformation and compensation

Figure 24 displays results obtained by using compensation techniques as described above. In the case of the level of deformation is again $\lambda/2$. The sources S1 and S3 (top and bottom) are supposed to be known, and we want to locate the unknown source S2 (in the middle). The localization of S1 and S3 is carried out only to check that we find the good

directions: during the experiments (in chamber or in flight), that enables us to check the effectiveness of both the compensation method and the calibration of the recorded signals.

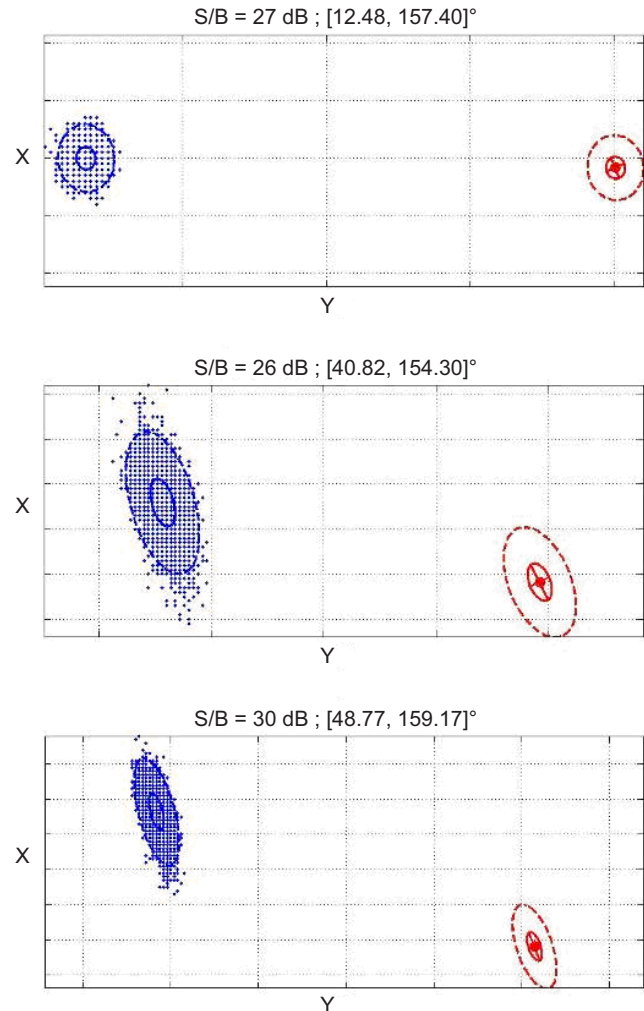


Figure 23 2D locations on ground of the 3 sources (with static deformation, amplitude $\lambda/2$).

Compare the results related to the source S2 in Figure 24 with those of Figure 23 and those of the reference case (Figure 22). Then, we observe the compensation capacity of the method, which is very encouraging for the continuation of our work. However, we observe a low bias, but which remains weak compared to the precision obtained.

This bias on S2 is also present on S1 and S3, which indicates it may be due to a calibration problem. This is not astonishing because, for the Ka band, the relative measures of location between the sources and the antennas must be extremely precise: an error of 1 mm may generate a phase error about 18° .

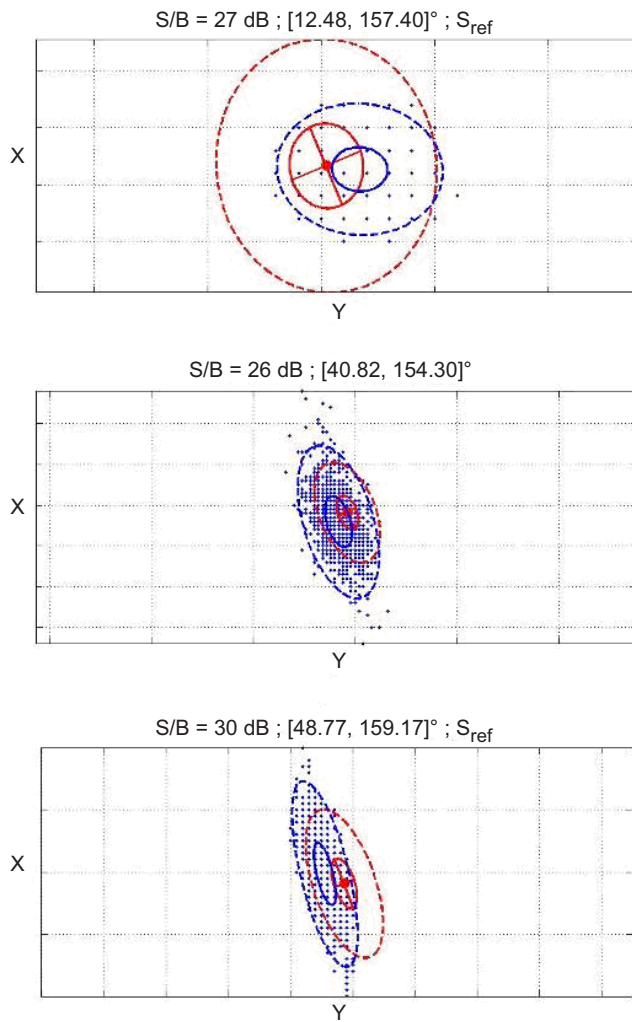


Figure 24 2D locations on ground with compensation and deformation; top and bottom sources are used as known sources to estimate deformation

7. CONCLUSIONS

In the present paper it has been shown that development of array antennas on load bearing aerospace structures has to take into account the effects of deformation and vibrations. The effects have been investigated for an array antenna on a vibrating plate. It was observed that the direction of the main beam of the generic radar antenna becomes non stationary. This negative effect can be suppressed by means of adaptive synthetic beam forming. To this end the phase differences between the antenna elements on the deformed and undistorted structure have to be determined instantaneously. This requires the measurement of out-of-plane variations by appropriate sensors, or the electronic measurement of phase changes.

Before setting up such an interferometer array on the motorglider, ONERA has performed a feasibility study with the aim of investigating techniques for compensating effects

of deformation and vibration of the wing. The study has addressed theoretical calculations, data-processing simulations and experimentation, on reduced scale, in an anechoic chamber. In this paper the results achieved so far have been described.

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BIOGRAPHY

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